Home Network Playbook

Maximizing home infrastructure: document it as found, then improve it. October 2026.

The pitch

Most homes already contain structured wiring that goes unused. This house has **Cat6e and RG6 in every room**, all terminating at one comm panel, yet the network reaches remote rooms over MoCA (Ethernet over coax) through a single splitter. This playbook records the setup so others can replicate it and lays out a measured path to use the wiring properly.

Phase 1: As found

Layer	What is there
WAN	AT&T Fiber to AT&T device (ONT) on exterior wall; AT&T Cat6 to comm panel
Panel	AT&T Cat6 patched to home Cat6e run to media room; RG6 splitter for remote rooms
Media room	AT&T router, then Nest mesh primary hub, then TP-Link switch, then Actiontec MoCA onto RG6
Remote rooms	Top-floor spare, main-floor sitting room, main-floor master: Actiontec MoCA, Cat6 patch, Nest mesh point, devices
Unused	Cat6e in every room (POTS-terminated); all other RG6 legs

Signal path

Fiber → ONT → panel → media room: **AT&T router** → **Nest hub** → **TP-Link switch** → **MoCA** → RG6 → panel splitter → **3 x (MoCA** → **Nest point)**

Phase 2: Improvement opportunities

ID	Opportunity	Effort	Impact
P2-1	Re-terminate room Cat6e from POTS plugs to RJ45 keystones (POTS uses only 2-4 of 8 conductors)	Medium	Very high
P2-2	Make the comm panel the core: patch panel + switch, one port per room	Medium	Very high
P2-3	MoCA-clean coax: 5-1675 MHz splitter, 75-ohm terminators, good connectors	Low	High
P2-4	Remove double NAT: AT&T IP Passthrough to Nest hub	Low	High
P2-5	Wired backhaul for Nest points over Cat6	Low	High
P2-6/7	Label and document everything; UPS for ONT, gateway, switch	Low	Medium

Measure, do not assume

Baseline every room with iperf3 before changes (expect about 940 Mbps over gigabit Ethernet), confirm double NAT with a traceroute, read MoCA link rates per node, and confirm mesh points report wired backhaul. Rollback is simple at each step because nothing is cut; old plugs and MoCA adapters stay until the Cat6 path is proven.

Replicate it

Full schematics (Mermaid), port map, step-by-step procedures, inventory template and glossary are in the repository docs: **01-as-found**, **02-improvements**, **03-inventory-template**, **04-verification**, **05-glossary**.